

FREDRIK PETTIT

(778) 903-6023 fredrikpettit@gmail.com linkedin.com/in/fredrikpettit github.com/freddyp3 fpettit-portfolio.vercel.app/

Education

University of British Columbia

Expected Graduation: May 2028

B.Sc. Computer Science (Artificial Intelligence) / GPA: 3.8

Vancouver, BC

Work Experience

AI Engineer Intern

Apr 2026 – Aug 2026

AEM

Langford, BC

- Scoped and shipped an AI customer support agent that resolves **42%** of tickets on its own for customers in 4 countries, at \$0.03 per resolved ticket versus \$8 of staff time, saving about **\$16K** a year and clearing an 800+ ticket backlog
- Traced most escalations to German and acronym queries, lifting auto-resolution 8 points once fixed, then built a 300-case regression eval from live conversations that caught a 6-point drop from an upstream model update within 4 hours
- Engineered a RAG pipeline over 4,000 pages of bilingual docs, with query rewriting feeding hybrid vector-keyword search on Azure AI Search and a reranker fine-tuned in PyTorch on 2,000 resolved tickets that lifted recall@5 from **77% to 87%**
- Cut monthly budget categorization from **6+ hours to 5 minutes** with a LangGraph agent that fixes vendor and accounting codes on 1,000 transactions, filtering most with rules and a cache before Claude, sending low-confidence fixes to a human

Software Engineering Intern

Jan 2026 – Apr 2026

ELLUSTROS

The Blue Mountains, ON

- Built the recommendation engine behind the company's chair-fitting app, matching each user's body measurements to the best of 100+ office chairs through layered filtering and weighted scoring across 15 ergonomic attributes, served via FastAPI
- Cut validation defects per sprint by **32%** in the company's chair-fitting mobile app by writing 40+ Jest and React Testing Library tests covering user sign-in flows and AI-driven measurement capture, run in a CI/CD pipeline on each merge

Teaching Assistant | CPSC 310, CPSC 110

Aug 2025 – Present

UBC Department of Computer Science

Vancouver, BC

- Supported 900+ students in UBC's largest computer science course, achieving a 4.9/5 average student evaluation score
- Wrote 250+ randomized JavaScript questions and Python autograders for CPSC 310 - Intro to Software Engineering

Projects

YouTube Shorts AI Optimizer | *Python, Claude API, MCP, Gemini, Google Cloud Run*

- Built an AI copilot that critiques draft YouTube Shorts before they are published and flags where viewers will swipe away, growing a personal channel from 1.2K to **38K** subscribers with **45M** total views, then launched it publicly to **100+** users
- Designed a multimodal eval scoring Gemini's video and audio analysis against retention on 85 held-out Shorts, reaching 0.45 Spearman correlation versus 0.25 for a veteran creator and pinpointing drop-off within 1.5s on 65% of them
- Shipped a remote MCP server so creators can get a publish plan without leaving Claude, running a tool-calling agent on Cloud Run that cut p95 latency from 95s to 18s with parallel tool calls and blocked **98.5%** of 200 prompt-injection attacks

Scheming - HackMIT 2026 | *Python, Claude, OpenAI, Modal, MongoDB*

- Built a lie detector for AI agents that flags when an agent's words contradict its own private reasoning, citing the exact line as evidence, and proved it on agents playing Werewolf, matching human judgment on **94%** of 300 hand-labeled turns
- Pointed the same judge at 1,500 real coding-agent sessions, flagging claims like "all tests pass" that the agent's own logs contradicted in 6% of runs at 92% precision, so teams can catch agents misreporting their work before merging

GitIntent - UC Berkeley AI LLM Hackathon 2026 | *Python, TypeScript, FastAPI, Redis Stack, voyage-code-3*

- Built a shared memory layer for AI coding agents that records every change with its prompt, then recalls failed attempts before any agent repeats them, lifting performance **8.3%** over the same agent without memory across 36 DeepSWE runs
- Designed the hybrid retrieval engine on Redis Stack, fusing HNSW vector search over voyage-code-3 embeddings with BM25 keyword ranking through Reciprocal Rank Fusion and deduplicating matches by snapshot, reaching ~93% recall@5

Technical Skills

Languages & Databases: Java, Python, C++, C#, SQL, R, JavaScript/TypeScript, PostgreSQL, DynamoDB, Redis

AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, sentence-transformers, embeddings, hybrid retrieval, RAG

LLM & Agents: LangGraph, LangChain, LangSmith, Claude/OpenAI/Gemini APIs, fine-tuning, guardrails, evals

Frameworks: Spring Boot, FastAPI, React, React Native, Expo, JUnit, Jest, Playwright, Pydantic, pandas, NumPy

Cloud & DevOps: AWS (Lambda, Step Functions, SQS), GCP Cloud Run, Azure AI Search, Docker, Git, CI/CD